Citation-Network Cohesion Across Low- and High-Impact Journals: A Matched Author Analysis

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Abstract

We examine whether author citation-network structure differs between low- and high-impact journal strata after matching authors on subject and recent citation productivity. A fixed venue classification based on subject-specific, publication-weighted Eigenfactor score cutoffs is used to construct 9,431 matched Case–Control author pairs from Crossref records for 2020–2024. Citation weights are fractionalized by the complete author counts of both works, while graph endpoints require identified authors. Four cohesion and concentration measures are primary outcomes; four additional measures are secondary. Pairwise inference uses complete pairs for each outcome, two-sided paired Wilcoxon signed-rank tests, paired bootstrap intervals, sign-flip checks, and within-family multiplicity correction. A separate Control-referenced Isolation Forest screen identifies unusual author–subject profiles and describes qualifying connected components without inferring intent. All 4 primary signed-rank comparisons met the 5% BH threshold; only 1 had a nonzero median paired difference. The screen flagged 0.70% of eligible Case rows and 0.04% of Controls (16.14-fold enrichment). The design distinguishes outgoing author neighborhoods from the matched-author induced graph and treats unavailable measurements as missing. The results characterize associations between venue strata and citation structure; they do not establish coordination, manipulation, or misconduct.

Keywords: scientometrics; citation networks; matched design; network cohesion; journal impact; anomaly screening

1 Introduction

Citation indicators are widely used to summarize scholarly influence, allocate resources, and support evaluation. Their convenience also creates incentives to optimize what is measured, which can weaken the connection between an indicator and the quality it is intended to represent (Biagioli et al., 2019; Goodhart, 1984; Hicks et al., 2015). Excessive self-citation, coercive citation, and concentrated citation exchange are therefore important objects of measurement in their own right (Biagioli & Lippman, 2020; Van Noorden & Chawla, 2019; Wilhite & Fong, 2012).

Most evidence on anomalous citation structure is either journal-level or begins with groups already selected for unusual behavior. Journal-network studies can identify exceptional exchange patterns (Kojaku et al., 2021), and analyses of citation stacking show how journal-level citation kinetics can be distorted (Heneberg, 2016). Author-level concentration indicators, in turn, can reveal when a small set of collaborators accounts for a large share of an author's citation impact (Evdaimon, Ioannidis, et al., 2024). Less is known about whether interpretable author-network measures differ across journal-impact strata when authors are compared within subject and at similar levels of recent citation productivity.

This study addresses that gap with a matched author–subject design. Low-impact and high-impact refer only to strata defined by subject-specific, publication-weighted 25th and 75th percentile cutoffs of journal Eigenfactor scores. We use *Case* and *Control* as compact labels for those operational strata; neither label evaluates the scientific merit of an author or work.

The study asks three questions:

1. Do matched Case and Control authors differ in citation-network cohesion and outgoing citation concentration?

2. Are author–subject profiles that are unusual relative to Controls enriched among Cases?

3. What is the descriptive structure of connected components formed by flagged profiles, and how do fractional citation weights mix across the two tiers?

The questions are associational. Network closure may reflect many mechanisms, including legitimate topical specialization and collaboration. Accordingly, the anomaly procedure is presented as a screening device, and component labels describe graph position rather than motive or conduct.

2 Related Work

Research evaluation based on citation counts, the h-index, and journal indicators has long been accompanied by cautions about field differences and strategic responses (Garfield, 2006; Hicks et al., 2015; Hirsch, 2005). Self-citation is common and can be substantively appropriate, but extreme rates complicate comparisons (Aksnes, 2003; Van Noorden & Chawla, 2019). Coercive citation and metric gaming demonstrate that observed references can also respond to editorial and institutional incentives (Baccini et al., 2019; Wilhite & Fong, 2012).

Several strands of work motivate a network perspective. Kojaku et al. identify anomalous groups in journal citation networks by comparing observed patterns with network expectations (Kojaku et al., 2021). Heneberg examines the transition from journal self-citation to citation stacking through journal-level citation flows (Heneberg, 2016). These approaches establish the value of relational structure but operate at the venue level. At the author level, Evdaimon et al. propose indicators based on the concentration of citing authors and hyper-collaboration (Evdaimon, Ioannidis, et al., 2024). Their concentration measures identify a narrow citing base; local closure, reciprocity, and concentration of an author's own references provide complementary information about how that base is connected.

The present analysis combines this network perspective with within-subject author matching. It does not attempt to classify behavior as legitimate or illegitimate. Instead, it estimates matched differences in a small, declared set of network measures and applies a Control-referenced screen whose output remains explicitly descriptive.

3 Data and Cohort

3.1 Source data and study window

The source is a fixed Crossref snapshot processed with *alexandria3k* (Spinellis, 2023). The study window covers citing and cited works dated 2020–2024. Available fields include Digital Object Identifiers (DOIs), publication year, normalized journal International Standard Serial Numbers (ISSNs), work references, and author rows with optional ORCID researcher identifiers. Crossref coverage and ORCID availability are incomplete; these limitations affect which citation endpoints can enter an author graph (Section 10).

Journals are assigned to Health Sciences, Life Sciences, Physical Sciences, Social Sciences and Humanities, or Multidisciplinary. The existing classification uses a deterministic, cached large-language-model annotation of representative journal content.

The retained venue classification uses fixed subject-specific journal scores from a separate directed journal-citation calculation (Bergstrom et al., 2008). Informally, the calculation repeatedly passes influence from citing to cited journals, while journal article counts supply the restart distribution. In the preserved implementation, let $Z_{jk}^{(s)}$ count citations from citing journal j to cited journal k in subject s. The diagonal is removed, and each non-dangling citing-journal column of H_s is defined by

$$(H_s)_{kj} = \frac{Z^{(s)}_{jk}}{\sum_\ell Z^{(s)}_{j\ell}}. \tag{1}$$

Let a_s be the normalized journal article-count vector and d_s indicate dangling columns. With damping $\alpha = 0.85$, power iteration applies

$$\pi_s = \alpha H_s \pi_s + \left[\alpha d_s^{\mathsf{T}} \pi_s + (1-\alpha)\right] a_s. \tag{2}$$

The received-influence vector $H_s \pi_s$ is normalized to sum to 100 within subject. Thus citation mass flows from citing to cited journals, journal self-citations are excluded, and both dangling mass and the 0.15 teleportation term follow article counts. These journal-level scores are an upstream exposure and are not computed on an author graph or used as an author-network outcome.

The current workflow consumes the fixed database scores as an upstream exposure. The intermediate journal-edge and article-count exports are unavailable, so this revision cannot independently reproduce or verify the score calculation end to end, including its extraction window. After scores are attached to eligible 2020–2024 work rows, the 25th and 75th percentile cutoffs are calculated across positive-score work rows within each subject. A journal therefore contributes once per eligible work: the cutoffs are publication weighted, not percentiles over distinct journals. Journals at or below the lower cutoff form the low-impact stratum and those at or above the upper cutoff form the high-impact stratum. No new venue classification was undertaken for this revision.

3.2 Author–subject cohort

The unit used throughout is an *author–subject*, identified by (ORCID, s). An author is a Case in subject s when at least 70% of eligible works in that subject appear in low-impact journals, and a Control when at least 70% appear in high-impact journals. Authors not meeting either rule are excluded from matching. A uniqueness check prevents more than one tier assignment for the same (ORCID, s) key.

The retained primary cohort contains 9,431 matched Case–Control pairs across the five subjects. The deterministic 10% ORCID sample and the original matching cohort are retained to separate analytical corrections from cohort redefinition.

4 Matching

Recent subject-specific citation productivity is summarized by h_5, the Hirsch index calculated from an author's works in subject s published during 2020–2024 (Hirsch, 2005). It is the largest integer h for which at least h of those works are each referenced at least h times in the fixed Crossref snapshot; mathematically, it can be zero. The retained upstream h_5 table materializes only positive values, so this fixed-cohort revision restricts the matching pool to $h_5 \geq 1$ and excludes otherwise tier-eligible zero-h_5 profiles before candidate generation.

Within the positive-h_5 pool, a Case–Control candidate is eligible only when $|h_{5,\text{Case}} - h_{5,\text{Control}}| \leq 3$. Eligible candidates are traversed in the deterministic total order subject, ascending absolute h_5 distance, Case ORCID, and Control ORCID. The first candidate whose two author–subject keys are both unused is retained, giving greedy matching without replacement within subject. Before an existing cohort can be reused, both positive h_5 records must be present, every pair must satisfy the inclusive hard caliper, and no author–subject key may be reused.

An immutable, read-only audit reconstructs the greedy match from the author profiles and materialized positive h_5 values, then compares row order, membership, and pair-set hashes with the retained table. Within that restricted pool, the direct hard-caliper recomputation reproduced the retained cohort exactly. The audit therefore verifies that replacing the bucket rule did not change pair membership within the implemented pool; it does not evaluate a rematch that admits $h_5 = 0$. Its complete output is written to

Table 1: Matching balance overall and by subject.

Subject	Pairs	h_5 observed	Exact h_5	Median Δh_5	Median $\lvert\Delta h_5\rvert$	Maximum $\lvert\Delta h_5\rvert$
Overall	9,431	9,431	5,736	0.00	0.00	2
1	3,776	3,776	1,651	-1.00	1.00	2
2	2,543	2,543	1,319	0.00	0.00	2
3	332	332	290	0.00	0.00	1
4	2,262	2,262	1,958	0.00	0.00	1
5	518	518	518	0.00	0.00	0

Δh_5 is Case minus Control. Every pair must satisfy $|\Delta h_5| \leq 3$. Exact-h_5 pairs form a sensitivity cohort; they do not replace the primary matched cohort.

`results/revision-v1/matching_audit.txt`.

The matching table supplies the Case ORCID, Control ORCID, subject, and both h_5 values. Feature rows are linked back to this table twice, on $(\text{case\_orcid}, s)$ and $(\text{control\_orcid}, s)$; ORCID-only joins are not used.

We report the distribution of h_5 and the within-pair difference before outcome testing. Because residual differences can remain under caliper matching, the four primary comparisons are repeated on the subset with exact h_5 equality.

The generated cohort audit found a maximum observed absolute h_5 difference of 2 and 0 pairs beyond the three-point caliper.

5 Network Construction and Definitions

5.1 Reference resolution and fractional weights

A citing-work/cited-work DOI pair is deduplicated before either work is expanded to author rows. Let n_c and n_a be the complete counts of author rows on the citing and cited work. These denominators include authors without ORCIDs. A resolved work citation contributes

$$\frac{1}{n_c n_a} \tag{3}$$

to each identified citing-author/identified cited-author combination. Rows with an unidentified endpoint do not produce analytic author edges; in particular, missing identifiers cannot become a shared pseudo-author. Multiple matched authors on one citing work do not multiply the underlying resolved work reference.

Every analytic edge is keyed by subject, citing ORCID, cited ORCID, and citation year. Only citing works assigned to the matched author's subject contribute to that author–subject's measures. Duplicate rows are aggregated so that this key is unique.

For coauthor classification, let c_{ij} be the first year in which authors i and j published together. A citation from i to j in year t is a prior-coauthor citation only if $c_{ij} < t$. A sensitivity analysis uses $c_{ij} \leq t$ to show the effect of admitting same-year collaborations.

5.2 Graph and reference populations

The analysis deliberately separates three analytic populations.

Outgoing ego layer. For each matched author–subject i, this layer contains citations from i to any identified author reached from an eligible citing work. It defines self-citation, prior-coauthor citation, outgoing concentration, and annual dyadic surge share.

Matched-author induced graph. For each subject s, this graph retains only nodes in the matched cohort for s and the edges among them. It defines reciprocity, local clustering, within-sample citation balance, outlier components, and tier mixing.

Deduplicated work-reference layer. This layer precedes author expansion and defines journal endogamy without requiring an identified cited-author endpoint.

The separation prevents a metric computed on all identified recipients or work references from being described as though it were computed only among matched authors.

5.3 Notation, metrics, and missing values

Let x_{ijt} denote the fractional citation weight from author i to author j in year t, after the construction above. Missing dyad–years are zero, and $w_{ij} = \sum_{t=2020}^{2024} x_{ijt}$ is cumulative weight. If V_s is the set of matched authors in subject s, the matched-author graph is the directed, weighted, loopless graph

$$G_s = (V_s, E_s, w), \qquad E_s = \{(i, j) : i, j \in V_s,\ i \neq j,\ w_{ij} > 0\}. \tag{4}$$

Its simple binary undirected projection is

$$U_s = (V_s, \widetilde{E}_s), \qquad \{i, j\} \in \widetilde{E}_s \iff i \neq j \text{ and } w_{ij} + w_{ji} > 0. \tag{5}$$

Direction and fractional weight are discarded only in U_s. Self-edges remain available in the outgoing ego layer for self-citation and surge share but do not enter G_s or its peer-network measures.

The study-specific temporal measure is

$$B_i = \frac{\max_{j,t=2021,\ldots,2024}[x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}. \tag{6}$$

Unavailable values remain missing through feature assembly. A zero is used only when the denominator and graph population are observed but the relevant event is absent. Thus a defined rate with no qualifying citations is zero; reciprocity is zero when internal outgoing weight exists but none is returned; clustering is zero for an observed induced-graph node with fewer than two neighbors; and journal endogamy is zero when resolved references exist but none remains within the citing journal. Outgoing HHI is missing when there is no identified non-self outgoing weight, balance is missing when total internal incident weight is zero, journal endogamy is missing when no resolved outgoing work reference exists, and surge share is missing when total identified outgoing weight is zero. These rules imply $0 \leq B_i \leq 1$.

Table 2: Declared author–subject measures.

Measure	Definition	Population and interpretation
Prior-coauthor citation rate	$\dfrac{\sum_{t,j\neq i} x_{ijt}\mathbf{1}(c_{ij} < t)}{\sum_{t,j\neq i} x_{ijt}}$	Outgoing ego layer. Fraction of identified non-self outgoing weight directed to an author whose first collaboration preceded the citation year.
Self-citation rate	$\dfrac{\sum_t x_{iit}}{\sum_{t,j} x_{ijt}}$	Outgoing ego layer. Fraction of identified outgoing weight assigned to the same ORCID.
Reciprocity	$\dfrac{\sum_{j\neq i} \min(w_{ij}, w_{ji})}{\sum_{j\neq i} w_{ij}}$	Directed, weighted G_s. Fraction of internal outgoing weight matched by reverse cumulative weight, capped dyad by dyad.
Local clustering	$\dfrac{2T_i}{k_i(k_i-1)}$	Binary projection U_s; T_i is the number of ties among i's k_i neighbors.
Outgoing Herfindahl–Hirschman index (HHI)	$\sum_{j\neq i} p_{ij}^2$, $p_{ij} = w_{ij}/\sum_{k\neq i} w_{ik}$	Outgoing ego layer. Concentration of identified non-self outgoing weight across recipients (Hirschman, 1964).
Journal endogamy	Resolved outgoing work references whose citing and cited journals have the same nonblank normalized ISSN, divided by all resolved outgoing work references	Deduplicated work-reference layer for the author–subject's citing works. This descriptive term does not imply impropriety.
Within-sample citation balance	$\dfrac{\sum_{j\neq i} w_{ij} - \sum_{j\neq i} w_{ji}}{\sum_{j\neq i} w_{ij} + \sum_{j\neq i} w_{ji}}$	Directed, weighted G_s. Positive values denote net giving and negative values net receiving within the sample.
Maximum annual dyadic surge share	$\dfrac{\max\limits_{\substack{j\\ t=2021,\ldots,2024}} [x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}$	Outgoing ego layer. Largest positive annual dyad increment as a share of total identified outgoing weight.

Notes. x_{ijt} is fractional weight from i to j in year t; $w_{ij} = \sum_{t=2020}^{2024} x_{ijt}$; c_{ij} is their first joint-publication year; and $\mathbf{1}(\cdot)$ is the indicator function. G_s retains positive-weight non-self edges among matched authors, and U_s links a pair when either direction is positive. HHI equals one when all non-self weight goes to one recipient and decreases as weight is dispersed. Also, $[z]_+ = \max(z, 0)$. A positive observed denominator with no qualifying event gives zero. A zero denominator gives a missing value, except that clustering is zero for an observed node with fewer than two neighbors. The surge definition is study-specific and is not an implementation of a state-transition burst-detection model. All measures are calculated separately for each author–subject.

Statistical comparisons use complete-pair deletion separately for each metric rather than a common imputed matrix.

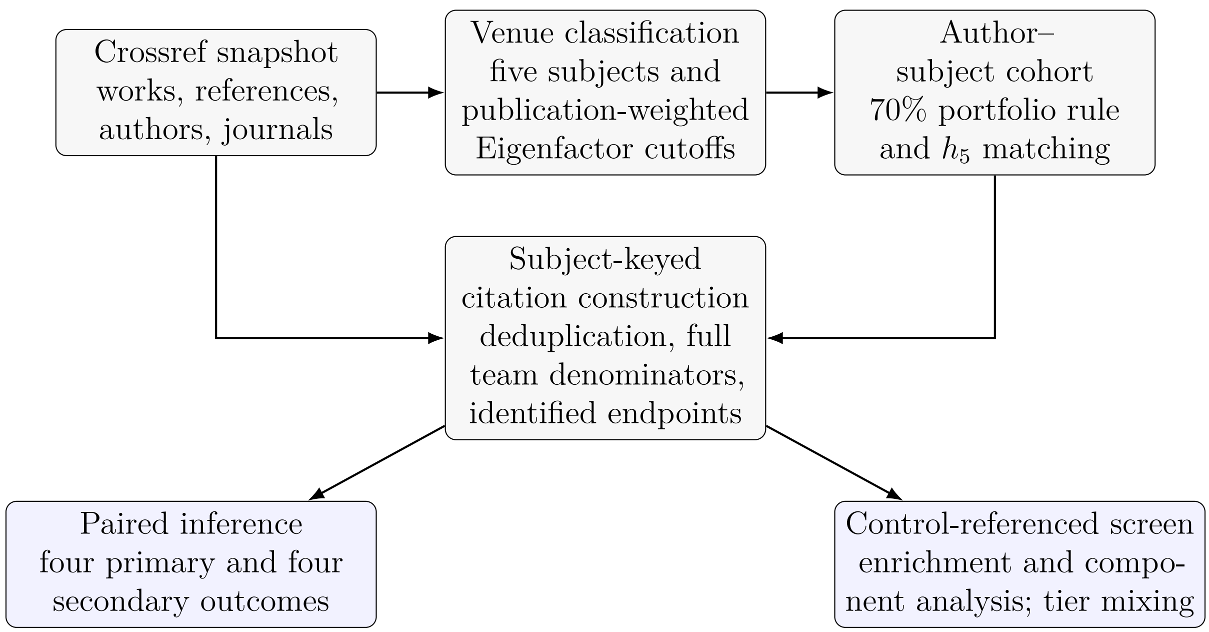

Figure 1: Reader-facing study design. Venue classification and author matching define the cohort; subject-keyed citation construction then supports two declared analytical branches.

6 Paired Inference

The four primary outcomes are prior-coauthor citation rate, reciprocity, local clustering, and outgoing HHI. The four secondary outcomes are self-citation rate, journal endogamy, within-sample citation balance, and maximum annual dyadic surge share. The former normalised burst-intensity feature is not an additional outcome; it was replaced by the surge share in Equation 6. The revised analysis therefore reports eight distinct outcomes in two declared families.

For outcome m, let $d_{pm} = x_{pm}^{\text{Case}} - x_{pm}^{\text{Control}}$ for matched pair p. Only pairs for which both values are available enter that outcome. Because several paired-difference distributions are bounded and tie-heavy, we use a two-sided paired Wilcoxon signed-rank test rather than a Gaussian paired t-test (Wilcoxon, 1945). We report the number of complete pairs, tier summaries, the median paired difference, and the matched rank-biserial effect size

$$r_{\text{rb}} = \frac{W^{+} - W^{-}}{W^{+} + W^{-}}, \tag{7}$$

where W^{+} and W^{-} are the positive and negative signed-rank sums after omitting zero differences. Ninety-five percent confidence intervals for the median paired difference are obtained from 2 000 bootstrap resamples of matched pairs, preserving both members of each pair (Efron & Tibshirani, 1993). Benjamini–Hochberg correction is applied to the Wilcoxon p-values separately within the four primary and four secondary families (Benjamini & Hochberg, 1995). The effect r_{rb} ranges from -1 to 1, with positive values indicating higher ranks for Cases. Benjamini–Hochberg correction limits the expected false-discovery proportion within each declared family.

As a randomization check, each observed within-pair difference is independently multiplied by $+1$ or -1 in 10 000 draws, using the absolute mean paired difference as the two-sided statistic. This sign-flip null preserves pair magnitudes and replaces the earlier pooled-label permutation. Primary outcomes are also repeated for exact-h_5 pairs and under the same-year coauthor definition.

7 Anomaly Screen

The anomaly analysis answers a different question from paired inference: whether profiles that are unusual relative to the Control distribution occur more often among Cases. It is not trained to predict tier and is not evidence of intent.

An author–subject is eligible only when it has at least one identified non-self outgoing citation. The rebuilt screen applies an explicit structural rule: it includes the five author-edge measures representing self-reference, prior-collaborator citation, mutual exchange, neighborhood closure, and recipient concentration—self-citation rate, prior-coauthor citation rate, reciprocity, local clustering, and outgoing HHI. Journal endogamy is excluded because it is defined on work references rather than author edges; within-sample balance is a signed flow position that is unavailable without internal incident weight; and surge share is temporal rather than a static cohesion or concentration measure. This rule defines the revised screen; it is not presented as independent validation or as a feature set prespecified in the original analysis. No additional feature weights are applied. Scoring additionally requires all five inputs to be available; citation eligibility and detector completeness are reported separately. Unavailable feature values are not recoded as structural zeros. Within each subject, eligible detector-complete Controls supply the feature medians and interquartile ranges. A nonfinite or zero Control interquartile range is replaced by 1 to avoid division by zero; the centered feature then remains in its original scale. The resulting transformation is also applied to Cases.

Each of 200 Isolation Forest trees recursively chooses a feature and split at random, using at most $\min(256, n_{\text{Control}})$ Controls per tree (F. T. Liu et al., 2008). Profiles isolated in fewer splits receive lower library scores; we negate `score_samples` so that larger study scores mean greater departure from the subject-specific Control distribution rather than a prediction of tier or conduct. One model is fitted per subject with seed 42. Although `contamination=auto` is passed to the library, its built-in decision cutoff is not used for the reported flag. A row instead passes the score criterion when its study score is above the empirical 99th percentile among Controls in that subject. This stringent tail operating point makes no Gaussian assumption and is not treated as uniquely optimal; the sensitivity analysis below also evaluates the 97.5th and 99.5th percentiles. A row passes the cohesion restriction when at least two of prior-coauthor citation rate, reciprocity, clustering, and outgoing HHI exceed their respective percentile cutoffs in the same Control reference set. Because all four of these measures also enter the detector, this is a deliberately overlapping, interpretable restriction rather than independent corroboration. The final flag is the intersection of the two criteria and is the only flag used in enrichment, component, and figure generation.

Sensitivity analyses cross the 97.5th, 99th, and 99.5th Control-percentile cutoffs with ten fixed random seeds. We also report the detector-by-restriction 2×2 table and Spearman rank association between detector score and cohesion exceedance count among detector-complete rows. A leave-one-feature-out analysis repeats the detector at the canonical threshold and seed while holding the canonical complete-row population and the four-measure restriction fixed. We report the proportion flagged in each tier, the Case share among flagged profiles, Case-versus-Control enrichment, and flag-set Jaccard similarity. These quantities describe an operational screen; they are not measures of classification accuracy because tier is not a ground-truth anomaly label.

For the final flagged-node set F_s, component analysis uses the directed weighted subgraph $G_s[F_s]$. Component membership is weak: direction is discarded only to find connected sets. Within each reported component, node flow is $\sum_j w_{ij} - \sum_j w_{ji}$. Directed normalized betweenness measures how often a node lies on weighted shortest paths, using edge length $1/w_{ij}$ so greater citation weight represents a shorter connection. Flow roles are described neutrally as *net giver*, *net receiver*, or *balanced flow*; the highest-betweenness node is identified separately. Five nodes is a reporting gate, not a topology threshold, and no topology label is assigned. A component figure is generated only when at least one component passes that gate; no illustrative or synthetic network is substituted.

Finally, a directed 2×2 mixing matrix is computed from G_s by summing fractional citation weight with citing tier in rows and cited tier in columns. If e is the matrix normalized to sum to one, with row and column marginals a and b, respectively, weighted nominal assortativity is

$$r = \frac{\mathrm{tr}(e) - a^\intercal b}{1 - a^\intercal b}. \tag{8}$$

Positive r indicates more within-tier weight than implied by the row and column marginals. The reported two-sided label-swap p-value compares the observed within-tier weight share, $\mathrm{tr}(e)$, with its distribution under 10 000 independent within-pair tier-label swaps; assortativity is reported descriptively and is not the permutation statistic. The swaps preserve the directed weighted graph and matching structure while breaking the observed association between tier and node identity.

8 Results

All numerical values in this section are generated by the canonical analysis under `results/revision-v1`. The manuscript does not search the historical `latest/`, `analysis_output/`, or `analysis_results_v4/` directories. If a canonical artifact is absent, the draft displays a pending marker rather than an earlier estimate.

8.1 Matched differences in network measures

All four primary signed-rank comparisons remained different after BH correction. Outgoing HHI was higher for Cases (median paired difference 0.035, 95% bootstrap CI [0.033, 0.038]; $r_{\mathrm{rb}} = 0.580$). Coauthor-citation rate, reciprocity, and clustering had zero median paired differences because of ties, although their matched rank-biserial effects were positive (0.116, 0.468, 0.395).

Table 3: Primary matched comparisons of citation-network cohesion.

Metric	n	Case med.	Control med.	Δ med.	Paired boot. 95% CI	r_{rb}	p_{BH}	p_{flip}
Coauthor-citation rate	6,848	0.000	0.029	0.000	[0.000, 0.000]	0.116	< 0.001	< 0.001
Reciprocity	464	0.000	0.000	0.000	[0.000, 0.000]	0.468	< 0.001	< 0.001
Local clustering	9,431	0.000	0.000	0.000	[0.000, 0.000]	0.395	< 0.001	< 0.001
Outgoing HHI	6,848	0.073	0.030	0.035	[0.033, 0.038]	0.580	< 0.001	< 0.001

Complete-pair deletion is performed separately for each metric. Positive differences and rank-biserial effects indicate Case > Control. BH correction is within the displayed outcome family.

Table 4: Secondary matched comparisons.

Metric	n	Case med.	Control med.	Δ med.	Paired boot. 95% CI	r_{rb}	p_{BH}	p_{flip}
Self-citation rate	6,908	0.000	0.007	0.000	[0.000, 0.000]	0.163	< 0.001	< 0.001
Journal endogamy	7,296	0.007	0.065	-0.028	[-0.030, -0.026]	-0.244	< 0.001	< 0.001
Within-sample citation balance	1,075	0.014	-0.259	0.000	[0.000, 0.000]	0.105	0.008	0.007
Maximum annual dyadic surge share	6,908	0.121	0.068	0.047	[0.044, 0.051]	0.514	< 0.001	< 0.001

Complete-pair deletion is performed separately for each metric. Positive differences and rank-biserial effects indicate Case > Control. BH correction is within the displayed outcome family.

Among secondary outcomes, journal endogamy was lower for Cases (median paired difference -0.028, 95% bootstrap CI [-0.030, -0.026]), whereas maximum annual dyadic surge share was higher (difference 0.047, 95% CI [0.044, 0.051]).

Table 5: Exact-h_5 and coauthor-timing sensitivity analyses.

Metric	n	Case med.	Control med.	Δ med.	Paired boot. 95% CI	r_{rb}	p_{BH}	p_{flip}
Coauthor-citation rate	4,221	0.030	0.026	0.000	[0.000, 0.000]	0.273	< 0.001	< 0.001
Reciprocity	297	0.000	0.000	0.000	[0.000, 0.000]	0.511	< 0.001	< 0.001
Local clustering	5,736	0.000	0.000	0.000	[0.000, 0.000]	0.556	< 0.001	< 0.001
Outgoing HHI	4,221	0.071	0.030	0.034	[0.031, 0.037]	0.569	< 0.001	< 0.001
Coauthor-citation rate (same-year included)	6,848	0.066	0.081	0.000	[-0.003, 0.000]	0.102	< 0.001	< 0.001

Complete-pair deletion is metric-specific. The first four rows use the exact-h_5 cohort and BH correction across those primary outcomes; the same-year coauthor row uses the full cohort as a separate one-outcome sensitivity family.

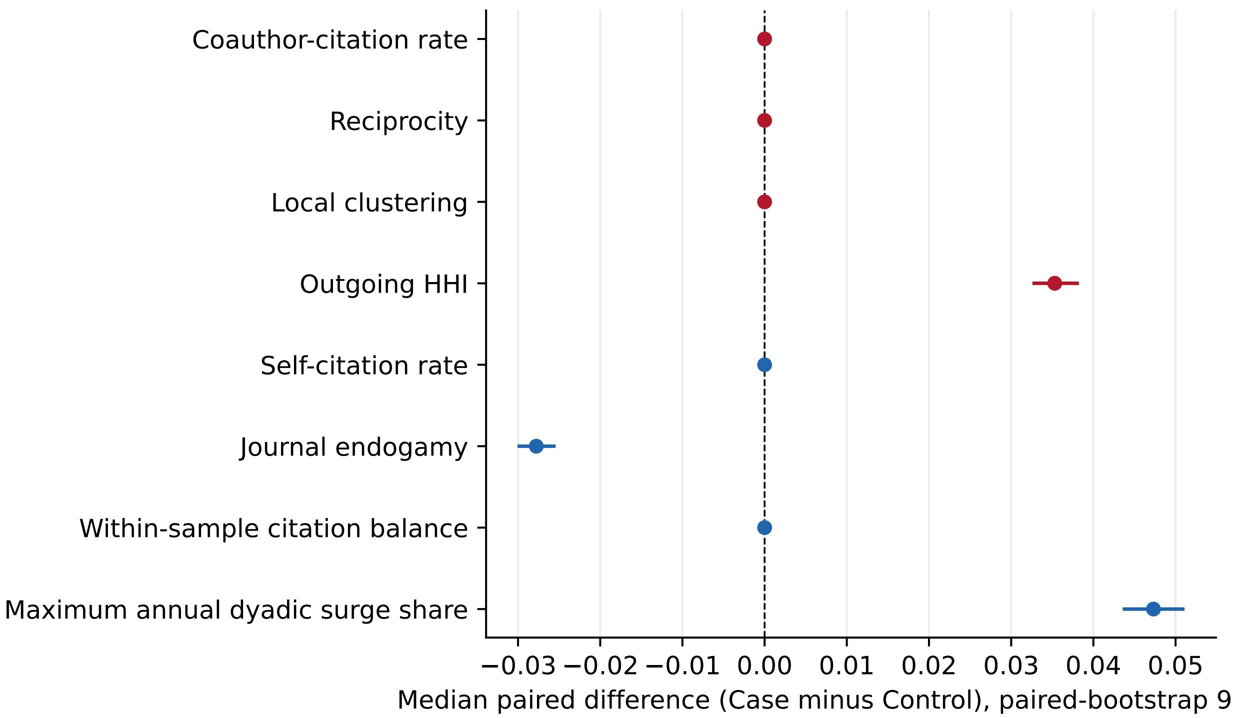

Figure 2: Matched effect estimates by outcome. Points show median paired differences and bars show paired-bootstrap 95% confidence intervals.

Pair counts are reported separately because the missingness pattern differs by measure. The count for every metric is checked not to exceed the 9,431 primary pairs.

8.2 Sensitivity of primary comparisons

All 4 exact-h_5 primary comparisons retained positive rank-biserial effects; exact-h_5 outgoing HHI had a median paired difference of 0.034 (95% CI [0.031, 0.037]). Under same-year

Table 6: Control-referenced anomaly-screen flags by journal-impact tier.

Tier	Eligible	Screenable	Flagged	Flagged share	Case/Control enrichment	Case share of flags	Fisher p
Case	6,960	1,897	49	0.70%	16.14	92.45%	< 0.001
Control	9,171	1,973	4	0.04%	16.14	–	< 0.001

Eligibility requires identified non-self outgoing citation weight. Flags are screening results and do not establish intent or misconduct.

Table 7: Overlap between detector-threshold exceedance and the cohesion restriction.

Detector exceeds threshold	No restriction	Restriction	Total
No	3,668 (94.78%)	2 (0.05%)	3,670 (94.83%)
Yes	147 (3.80%)	53 (1.37%)	200 (5.17%)
Total	3,815 (98.58%)	55 (1.42%)	3,870 (100.00%)

Cells are counts (percent of 3,870 screenable author–subject rows). Spearman's ρ between detector score and cohesion exceedance count was 0.385 ($n = 3,870$; $p < 0.001$). The four cohesion measures are detector inputs, so this is an overlapping interpretable restriction rather than independent corroboration.

coauthor classification, the median paired difference was 0.000 (95% CI [-0.003, 0.000]).

The sensitivity table is interpreted by direction, interval width, and effect magnitude, not solely by whether an adjusted p-value crosses a threshold. Any weakening, reversal, or loss of precision in the corrected analysis is retained in the generated output.

8.3 Control-referenced anomaly enrichment

Of 6,960 eligible Case and 9,171 eligible Control rows, 1,897 and 1,973, respectively, had all detector inputs. The canonical rule flagged 49 Cases and 4 Controls (0.70% versus 0.04% of eligible rows; 16.14-fold enrichment; $p < 0.001$ by Fisher's exact test).

Among 3,870 screenable author–subject rows, 200 exceeded the detector threshold, 55 met the cohesion restriction, and 53 met both; 147 were detector-only and 2 restriction-only. Detector score and cohesion exceedance count had Spearman $\rho = 0.385$ ($p < 0.001$).

Across 5 leave-one-feature-out detector fits, final-flag Jaccard similarity with the canonical flags ranged from 0.907 to 0.981, and Case/Control enrichment ranged from 15.15 to 16.80. Full leave-one-feature-out results appear in Table 11.

Table 8: Anomaly-screen threshold and seed sensitivity.

Control percentile	Flagged, median [range]	Case share	Control share	Enrichment	Jaccard vs. canonical
97.5%	97 [93–99]	1.21%	0.13%	9.22	0.55
99.0%	54 [52–55]	0.71%	0.04%	16.31	0.98
99.5%	42 [41–44]	0.55%	0.04%	12.68	0.77

Each row summarises ten fixed seeds; the canonical screen uses the 99th percentile and seed 42.

Table 9: Descriptive structure of outlier components with at least five nodes.

Subject	Component	Nodes	Dyads	Weight	Net givers	Net receivers	Highest betweenness
No component contained at least five flagged nodes.							

Components use only the canonical final flag and cumulative weighted dyads. Flow labels are descriptive and do not imply coordination or intent.

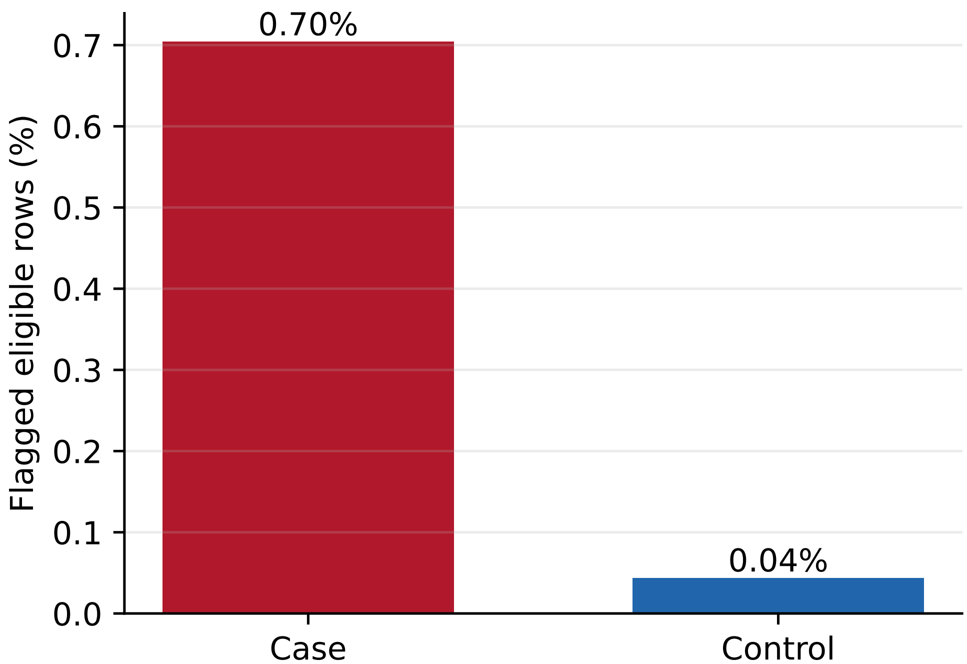

Figure 3: Share of eligible author–subjects flagged in each tier under the declared screen. Values and supporting counts are generated from the same final flag used in all downstream analyses.

8.4 Flagged components

No outlier component contained at least five flagged nodes.

No component figure is included in this build. The analysis emits one only if a component formed from the final flag contains at least five nodes.

Table 10: Weighted citation mixing among matched authors.

Citing tier	Cited Case	Cited Control
Case	226.87	14.78
Control	3.73	58.38

Cells are cumulative fractional citation weights. The within-tier share is 93.91%, weighted assortativity is 0.824, and the two-sided 10,000 within-pair tier-label-swap p-value is < 0.001.

8.5 Fractional tier mixing

Within-tier citations represented 93.91% of matched-sample weight (label-swap $p < 0.001$).

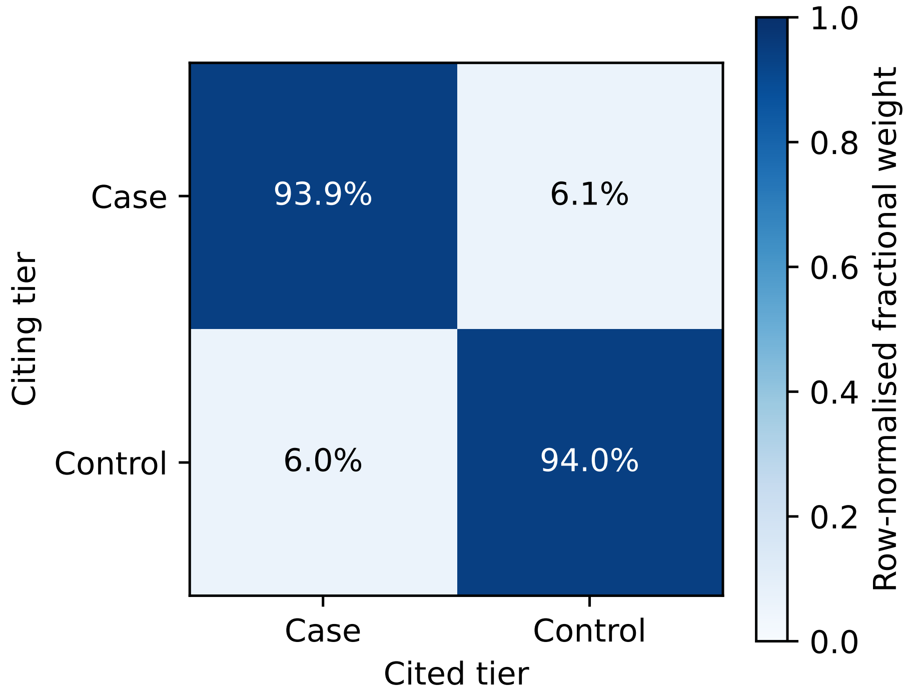

Figure 4: Fractional citation-weight mixing between Case and Control nodes. The comparison distribution is based on within-pair tier-label swaps.

9 Discussion

This matched design evaluates whether citation structure is associated with an operational journal-impact stratum after conditioning on subject and recent author citation productivity. The estimand is therefore a within-pair difference, not the effect of publishing in a particular venue. Journal choice, collaboration, topic, and citation behavior can influence one another, and the observational design does not order those mechanisms causally.

The regenerated evidence is summarized as follows. All 4 primary signed-rank comparisons met the 5% BH threshold; only 1 had a nonzero median paired difference. The screen flagged 0.70% of eligible Case rows and 0.04% of Controls (16.14-fold enrichment). No outlier component contained at least five flagged nodes. Within-tier citations represented 93.91% of matched-sample weight (label-swap $p < 0.001$). The detailed estimates, intervals, and sensitivity results in Section 8 govern interpretation when a short summary masks heterogeneous directions or uncertainty.

The only primary outcome with a nonzero median paired difference was outgoing HHI, indicating greater concentration of Cases' identified non-self outgoing citation weight across recipients. This is a directional complement to, rather than a replication of, Evdaimon et al.: their concentration indicator counts the incoming citing authors needed to account for 50% of an author's received citations, whereas outgoing HHI measures how an author distributes references (Evdaimon, Ioannidis, et al., 2024). The zero median paired differences for prior-coauthor citation rate, reciprocity, and local clustering, despite positive rank-biserial effects in their tie-heavy distributions, further show that recipient concentration did not translate into a uniform median shift in collaboration or closure. That separation is compatible with Evdaimon et al.'s finding that extremes on their concentration- and collaboration-based indicators only partly co-occur, but the different edge direction, sampling frame, and author populations preclude numerical comparison.

Among secondary outcomes, Cases had lower same-journal endogamy but a higher maximum annual dyadic surge share. The first result does not support a simple account of greater same-journal referencing among Cases, while the second identifies a larger concentration in the greatest positive year-to-year increment to one recipient. Neither is a replication of Heneberg's journal-level evidence: that study compared the timing of journal self-citations and foreign citations and documented citation stacking around the journal-impact-factor window, whereas the present measures concern work-reference ISSN equality and author dyads over a fixed five-year window (Heneberg, 2016).

The enrichment and mixing results also address a different claim from CIDRE. Kojaku et al. classify journal-to-journal edges as excessive against a null model accounting for journal size and scientific communities, form weakly connected residual groups, and compare those groups with JCR suspensions (Kojaku et al., 2021). Here, the 16.14-fold Case–Control difference is enrichment under a Control-referenced author-profile rule, and the 93.91% within-tier weight is compared with within-pair label swaps; weighted assortativity is a separate descriptive summary. Neither result establishes excessive citation exchange or validates a group detector. No flagged component met the declared five-node reporting gate, so this study offers no component topology to compare with CIDRE's journal groups; that result does not exclude smaller connected flagged sets.

Several non-adversarial mechanisms could produce differences in cohesion or any observed enrichment. Narrow specialties may have small reference communities. Geographic and language proximity may shape both collaboration and citation, especially where research questions are locally grounded. Stable teams naturally generate prior-coauthor references and closed neighborhoods. Differences in journal indexing, reference deposit practices, and ORCID adoption can also change the observable graph even when underlying behavior is similar. These explanations are alternatives to, and may coexist with, any strategic response to evaluation incentives.

The Control-referenced procedure should consequently be understood as triage. It identifies profiles that are unusual under a declared reference distribution and adds a transparent cohesion restriction. The two criteria deliberately share inputs, so their intersection is a guardrail rather than two independent lines of evidence. It does not estimate the probability of manipulation, determine whether a citation was intellectually warranted, or establish intent. Component structure and net-flow labels are summaries of observed edges only. Any research-integrity use would require independent validation from full texts, editorial records, author disambiguation, and domain experts.

10 Limitations

Crossref reference coverage varies by publisher, journal, year, and document type. Missing references remove edges, and missing ORCIDs remove analytic endpoints even though all author rows still enter team-size denominators. ORCID coverage may correlate with geography, career stage, or venue. The resulting author graph is therefore a coverage-dependent projection of the literature rather than a complete citation network.

The five broad subject categories cannot capture fine topical proximity. The journal classifier was not independently validated against blinded human coding, and the publication-weighted Eigenfactor cutoffs remain sensitive to the observed journal network. Missing intermediate journal-edge and article-count exports prevent an independent end-to-end reconstruction of the fixed upstream scores. The 70% portfolio rule, deterministic ORCID sample, and exclusion of otherwise tier-eligible zero-h_5 profiles further limit generalization. Matching on subject-specific positive h_5 improves comparability within that restricted pool but does not balance geography, language, institution, team structure, career age, or unmeasured research topic.

Graph metrics depend on population boundaries. Ego-layer measures include identified recipients outside the matched cohort, whereas reciprocity, clustering, balance, components, and mixing are restricted to matched nodes. Results from the latter should not be interpreted as full-network properties. The 2020–2024 citation window limits observed citation behavior, while incomplete historical work and author metadata in the snapshot can left-censor or misdate prior collaborations. The year dimension of the edge data supports coauthor timing and the scalar surge-share outcome, but is not analyzed as a tier-specific trajectory or tier-by-year interaction; we therefore make no claim that a Case–Control gap widens over time.

The anomaly thresholds and Isolation Forest define an operational screen, not a validated diagnostic. Percentile, seed, and leave-one-feature-out sensitivity quantify some model dependence but do not supply ground-truth labels or make the overlapping restriction independent. Finally, structural metadata cannot reveal the textual function, scientific relevance, or motivation of a citation (J. Liu et al., 2024).

11 Conclusion

Conditional on the fixed upstream venue scores and retained positive-h_5 cohort, this study provides an auditable downstream framework for comparing citation-network cohesion across journal-impact strata. It corrects work-level deduplication and team-size denominators, uses author–subject keys, separates ego and induced graph populations, and limits inference to eight interpretable measures. All 4 primary signed-rank comparisons met the 5% BH threshold; only 1 had a nonzero median paired difference. The screen flagged 0.70% of eligible Case rows and 0.04% of Controls (16.14-fold enrichment). No outlier component contained at least five flagged nodes. Within-tier citations represented 93.91% of matched-sample weight (label-swap $p < 0.001$). These findings should be read as associations under incomplete bibliographic coverage. The screening results can prioritize descriptive follow-up, but they cannot establish coordination, intent, or misconduct.

12 Author Contributions

Conceptualization: P.-A.S., D.S.; Methodology: P.-A.S.; Software: P.-A.S., G.A.; Data curation: P.-A.S., G.A.; Validation: P.-A.S., G.A., D.S.; Formal analysis: P.-A.S., G.A.;

Visualization: P.-A.S.; Writing—original draft: G.A.; Writing—review and editing: P.-A.S., G.A., D.S.; Supervision: D.S.; Project administration: D.S.

13 Competing Interests

The authors declare no competing interests.

14 Funding Information

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15 Data and Code Availability

The revised software, aggregate replication materials, and retained-cohort database are archived as Zenodo version 3.0.0 (Spanakis et al., 2026). The archive contains the subject-keyed offline workflow, SQL and Python tests, manuscript sources, the exact aggregate macros, tables, and figures consumed from `results/revision-v1`, and the ORCID-linked retained cohort and matching covariate. The canonical builder reads only `author_matched_pairs` and `author_subject_h5_index` from the released database; its other legacy derived tables are non-authoritative and their descriptive metrics are not findings of misconduct. The approximately 128 GB raw Crossref snapshot and unavailable upstream journal-edge and article-count exports are not included. Historical local output directories are non-authoritative and are not manuscript inputs.

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A Matching and Sensitivity Specifications

Cases and Controls require at least three eligible works and a 70% portfolio share in the relevant journal stratum. The retained matching pool additionally requires a materialized positive h_5 value; zero-h_5 profiles are not candidates. Within that pool, candidate eligibility uses the inclusive hard caliper $|h_{5,\mathrm{Case}} - h_{5,\mathrm{Control}}| \leq 3$. Greedy selection proceeds without replacement within subject in the total order subject, ascending absolute h_5 distance, Case ORCID, and Control ORCID. The retained cohort must pass the same subject-keyed positive-h_5 join and caliper checks before reuse. Exact-h_5 pairs form a sensitivity subset rather than a replacement cohort.

The paired bootstrap resamples pair identifiers with replacement. The sign-flip test acts on complete-pair differences for one metric at a time. Multiple-testing adjustment is not pooled across primary and secondary families. The same-year coauthor sensitivity changes only $c_{ij} < t$ to $c_{ij} \leq t$; all other definitions and pair membership remain fixed.

B Anomaly-Screen Specification

Controls define subject-specific robust centers, scales, detector fits, score cutoffs, and univariate cohesion cutoffs. Cases do not affect those reference quantities. The canonical feature output retains the detector score, cohesion-restriction indicator, citation eligibility, and final flag for

Table 11: Leave-one-feature-out anomaly-screen sensitivity at the canonical threshold and seed.

Detector feature omitted	Flags	Case flags	Control flags	Enrichment	Jaccard vs. canonical
Self-citation rate	53	49 (0.70%)	4 (0.04%)	16.14	0.963
Coauthor-citation rate	50	46 (0.66%)	4 (0.04%)	15.15	0.907
Reciprocity	52	48 (0.69%)	4 (0.04%)	15.81	0.981
Local clustering	55	51 (0.73%)	4 (0.04%)	16.80	0.964
Outgoing HHI	51	47 (0.68%)	4 (0.04%)	15.48	0.962

Each fit uses the 99.0% Control percentile and seed 42. The canonical detector-complete population and cohesion restriction are held fixed; Case and Control percentages retain eligible-row denominators.

each (ORCID, s, tier) row. The downstream component and mixing routines consume that final flag rather than recomputing or substituting a label.

The primary screen uses seed 42 and the 99th Control percentiles. The threshold analysis repeats the complete fit at the 97.5th, 99th, and 99.5th percentiles for ten fixed seeds. A row with an unavailable required detector input is not zero-imputed; it receives no detector score, cannot be flagged, and is identified by a separate detector-completeness field. The overlap table is restricted to rows on which both criteria are defined.

C Reproducibility Checks

The rebuild checks the following manuscript-facing invariants:

- (ORCID, s, tier) is unique in the feature output;
- retained pairs have complete subject-keyed h_5 records, satisfy the inclusive three-point caliper, and do not reuse an author–subject key;
- every analytic edge has non-null endpoints and is unique on (s, i, j, t);
- each per-metric complete-pair count is no greater than 9,431;
- every observed surge share lies in $[0, 1]$;

- the same final flag supplies every flag-dependent enrichment, component, and figure artifact;
- a component figure is absent unless a component has at least five nodes; and
- reported numerical values enter through generated LaTeX artifacts rather than hand-copied prose.

SQL fixtures separately test fractional weight, null endpoints, work-reference deduplication, subject retention, and prior-coauthor timing. Python unit tests cover matching eligibility and deterministic order, retained-cohort validation, metric formulas, paired inference, subject-keyed joins, detector overlap, feature ablation, weighted mixing, and flag reuse.

D Journal Subject Classification

The retained journal mapping was produced with OpenAI `gpt-4.1` (snapshot `gpt-4.1-2025-04-14`) at zero temperature from journal names and representative work titles and abstracts. Prompts and cached outputs are part of the replication materials. Because the mapping has no independent human-validation sample and no open-weight baseline, classification error remains a limitation rather than a measured source of uncertainty (LLM Guidelines Consortium, 2024).